Document-Image Perceptual Hashing for Content Authentication

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***Abstract*—This paper proposes an end-to-end two-branch network for a document image perceptual hashing scheme, where the two branches focus on image visual features and text features, respectively. Existing perceptual hashing schemes cannot solve the problem of the tiny proportion of text tampering detection, while simple text detection is unable to solve the problem of background region-aware matching. To address these issues, we extract text information via optical character recognition (OCR) and then generate the text features using the bidirectional encoder representations from Transformers (BERT). Visual features of the image are extracted from the local and global features of the image using ResNet and Vision Transformer cascades, and then fused to generate the final hash sequence through the fully connected layer. The proposed network considers both image visual features and textual information to verify that the document image has not been tampered with. In our network, the OCR module enables accurate and intelligent text detection and recognition, particularly for dealing with text tampering that has only been conducted in tiny portions. It also provides more efficient and robust text recognition services. Experimental results show that the proposed hashing scheme is robust and discriminative in document images.**

***Index Terms*—Perceptual image hashing, document image, tam pering detection, two-branch network.**

I. Introduction

With the development of multimedia computers and online social networks, information can be disseminated in multiple modalities, such as text, images, or videos. Images containing text are one of the most common media, combining both visual and textual information. Common text images include scanned documents, certificates, invoices, etc. In this paper, we broadly refer to captured or scanned images that contain text as document images. In modern society, document images are used in various application scenarios, such as online education and e-government file management, and have become an essential medium for information communication. However, with the increasing popularity and improved functionality of image editing tools, the credibility of crucial document images is sometimes questioned. Critical information in document images can be altered through various image processing operations [1], [2], [3], [4], such as inserting text through imitation, splicing, and copy-paste [5], [6]. In the daily scenario, forged official seals, letters, insurance policies, and photo IDs are commonplace and are even used for commercial fraud and to influence public opinion. The use of tampered screenshots of important documents such as contracts, qualification certificates, or transaction records may result in financial loss. In the context of document images, unauthorized, malicious, or misleading modifications to the content of document images can directly affect the accuracy of the information conveyed by the document. Ensuring authenticity and detecting forgeries is crucial to protect individual rights, maintain social order, and promote information security.



Fig. 1. A pair of Chinese document images that change some of the Chinese text. The orange boxes indicate some examples of images with tampered regions, and the tampering traces in each image are barely visible.

Forgery detection in document images presents unique challenges compared to natural images. In document images, the tampered region may be tiny and blends subtly with the background, as shown in Fig. 1, where some of the Chinese text has been modified to significantly alter the semantic information of the document images. This subtlety makes detecting forged text more challenging than detecting forgeries in natural images. Even with state-of-the-art generalized image forgery localization methods such as PSCCNet [7], identifying forged regions in text images is still challenging. Existing perceptual image hashing techniques [8], [9] play an important role in image content authentication; however, they focus mainly on the visual features of the image rather than the textual content. However, document images differ from general images in that the textual information they contain is the key to understanding the document's meaning. Pure text verification can generate

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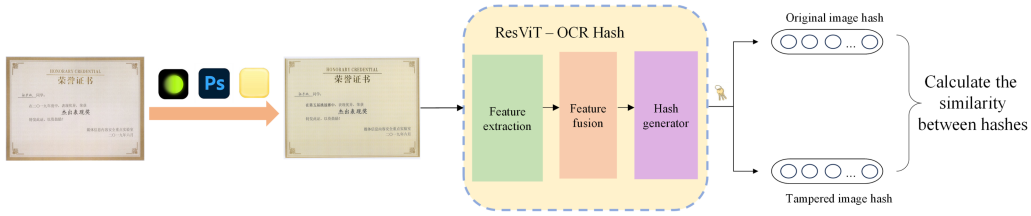


Fig. 2. Process of document image tampering authentication, including image tampering, feature extraction, hash generation, and tampering detection.

false positives or recognition errors due to the diversity or ambiguity of text. Document image content authentication not only needs to consider the visual features of the image, but more importantly, it needs to accurately identify and encode the text content in the image, providing a more comprehensive and accurate hash sequence for the document image. The existing perceptual hashing methods often struggle to accurately capture this critical information in text-rich document images, limiting their performance in detecting character-level tampering. The optical character recognition (OCR) technique improves data entry efficiency by automating text extraction from images. However, its accuracy is highly dependent on the clarity of the input image. For documents that are blurry, skewed, or contain complex layouts, the OCR technique may struggle to recognize them accurately. Therefore, an OCR-based document perceptual image hashing algorithm is proposed to provide an efficient and reliable content authentication scheme for document images. The text in the document image is comprehensively analyzed using the OCR technique to solve the problem of detecting character tampering. At the same time, the model's perceptual hash module extracts key visual features and effectively detects tampering in visual aspects such as background and masking. By combining the OCR system with perceptual hashing, the final generated hash sequence contains common features of both, and comprehensive authentication of document images is achieved, improving the reliability and efficiency of content authentication.

A new perceptual hash model for document images is shown in Fig. 2, which consists of three parts: (1) Feature extraction module, which is the first step to represent all modalities of the data. For image feature extractors, many approaches use feature extractors based on convolutional neural networks (CNNs) [10], [11], but small-scale features are often lost due to the down-sampling process. In addition, simple CNNs have difficulties in capturing global semantic interactions and contextual information. To improve the performance of deep learning models in image recognition and processing tasks, we combine ResNet and Vision Transformer (ViT) to extract visual features, employ an OCR module to extract textual content, and use the bidirectional encoder representations from Transformers (BERT) module to generate the textual feature map. (2) Feature fusion module, which performs a simple linear transformation of the textual feature map to match the image modality's feature vector dimensions, and then stitches the text feature vector and image feature vector into a text-image feature vector. (3) Hash module, which compresses the fused feature information into a fixed-length hash sequence. In order to achieve tampering detection of document images, we propose the ResViT-OCR network. The main contributions of this paper can be summarized in the following four aspects:

- To the best of our knowledge, this paper proposes for the first time that perceptual hashing combined with OCR technique can be used to tamper-proof content authentication in document images, thereby enriching existing image authentication systems.
- The image hashing module consisting of ResNet and ViT is used to extract image features. The convolutional layer of ResNet excels at capturing local features, while the residual learning framework enables deeper networks without sacrificing training efficiency. ViT's self-attention mechanism effectively handles global dependencies. This combined approach leverages ResNet's ability to capture image details at different scales and ViT's ability to process and understand global image content, resulting in more refined and comprehensive feature extraction.
- The text hash module extracts the textual information from document images, using the BERT module for the extraction of text features. After adding the OCR technique, the designed system not only captures the basic visual features of the image but also recognizes and processes the textual information, significantly enhancing the hash algorithm's sensitivity to textual information and the accuracy of the hash values.
- Our scheme efficiently extracts image and text features, integrates the feature extractor and hash generator in an end-to-end framework and demonstrates robust performance in document image authentication.

The rest of this paper is organized as follows. Section II reviews the related work on image hashing algorithms. Section III details our proposed hashing algorithm. Section IV presents comprehensive experiments, visualization analysis, and performance comparison. Finally, we summarize the whole work in Section V.

II. Related Work

A. Document Image Editing and Tampering

With deep learning, image editing algorithms can tamper with the textual information of an image in an end-to-end manner, thus posing a considerable security threat to document images. Wu et al. [3] introduced a network designed for text image editing that can replace text in scene text without destroying the image's inherent stylistic features. The technique first learns the style of foreground text and then transfers this style to new text

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provided by the user. Then, the newly styled text is merged with a texture that seamlessly matches the background of the original image. In another study, Roy et al. [12] proposed a character-level text editing network that modifies individual characters in a text image. Liu et al. [13] presented a technique that attempts to erase or conceal text within an image, favoring privacy protection. However, it may be exploited for nefarious purposes. The squeezing-excitation (SE) network proposed by Zhang et al. [14] is a multi-task SE network with two encoder-decoder streams, which utilizes image edges to learn the splicing mask. These advanced text editing techniques highlight the growing need for robust document image authentication methods.

B. Perceptual Image Hashing

According to the different feature extraction of image hashing, the traditional hashing schemes can be categorized into three types. Based on the spatial domain approach, Huang and Liu [15] extracted the maximum gradient and the corresponding orientation information from RGB color channels, and then further obtained the normalized histogram and local frequency coefficient features to generate the hash sequences. Su et al. [16] proposed a new personalized authentication framework, which incorporates the distance metric learning technique and the supervised quantization strategy into the image authentication process. Based on the transformed domain methods, they mainly include orthogonal and integral transforms to extract image features in the transform domain to generate hash. In [17], the image hash was generated based on the discrete cosine transform invariance at the same locations in separate blocks of the image. Tang et al. [18] proposed to present the edge image of the normalized image by Canny's algorithm and applied the 2D-discret wavelet transform (2D-DWT) to obtain the different sub-bands and calculate the weighted sum of the DWT of these sub-bands for generating the hash. Dimensionality reduction methods mainly utilize data dimensionality reduction algorithms to construct image hashes through low dimensional feature data, such as singular value decomposition [19], local linear embedding, nonnegative matrix factorization [20], and principal component analysis (PCA). Tang et al. [21] utilized quaternionic singular value decomposition to construct image hashes, improving the algorithm's uniqueness.

With the advancement of image processing technology and the development of deep learning, deep neural networks have been increasingly used in image hashing algorithms and have achieved better performance than traditional algorithms. Li et al. [22] applied deep learning methods to the field of perceptual hashing and demonstrated performance that outperformed traditional hashing algorithms. They proposed a robust and secure image hashing scheme based on deep neural networks, which utilized denoising self-coding to train to restore distorted images to their original state. Qin et al. [10] proposed a perceptual hashing algorithm based on a multi-constrained convolutional neural network using two constraints to model training and dynamically adjust the constraint values to improve perceptual robustness and discriminative ability. Liu et al. [23] proposed a Swin Transformer-based perceptual image hashing scheme to resist the effect of LCD screen distortion on content authentication; the scheme added a distortion simulator to assist the feature extractor and hash generator to learn and resist screen distortion. Yu et al. [24] proposed a robust hashing scheme with deep features and Meixner moments for image copy detection, extracting global deep features and local features separately. Hash is ultimately determined by quantifying and combining global depth features and local features.

The development of robust perceptual hashing methods for document images based on deep learning hashing technology provides a solution. The traditional method is based on artificial features, by comparing the histogram, edge features, and texture features of images to evaluate the similarity of documents. It is difficult to fully and accurately describe a document image's semantic and content characteristics because of the characteristics of manual design. The deep learning model can automatically learn the complex features in document images. Our work integrates visual and textual features to achieve comprehensive tampering detection and content authentication.

C. Perceptual Image Hashing versus Semantic Retrieval Hashing

Image hashing research primarily focuses on semantic image hashing retrieval and perceptual hashing authentication. Although both rely on hashing techniques, their goals and methods are somewhat different. Semantic retrieval hashing aims to find images semantically similar to the query image. For instance, Xing et al. [25] proposed deep incremental hashing, which uses a pointwise loss function to guide hash neural network learning and combines hash tables from different time steps via a weighted method to form a multi-table hash system. Nie et al. [26] introduced ssupervised discrete multiple-length hashin, enabling simultaneous learning of hash codes with multiple lengths, significantly improving retrieval flexibility and accuracy. Yang et al. [27] developed robust discrete hashing for large-scale image semantic indexing, which learns discrete binary codes and hash functions in a unified model. Lu et al. [28] proposed a subregion-guided transformer hash network for fine-grained image retrieval, generating compact binary codes through partial effective region erasure and multi-stage hybrid loss. Cao et al. [29] designed a privacy-preserving image retrieval framework using cycle-consistent adversarial networks and visual graph convolutional network to enhance retrieval accuracy. While semantic image hashing methods perform well in capturing high-level semantic information for efficient image retrieval, they are not directly applicable to document image authentication tasks.

D. Bidirectional Encoder Representations From Transformers

Text features need to be extracted first to fuse text and image information. BERT is a pre-trained language representation model proposed by Google [30] in 2018, which has shown great success in a wide range. The core of the BERT model is based on the Transformer architecture, which is capable of understanding and generating complex nuances and contextual dependencies in text through deep bi-directional learning. The pre-training of the

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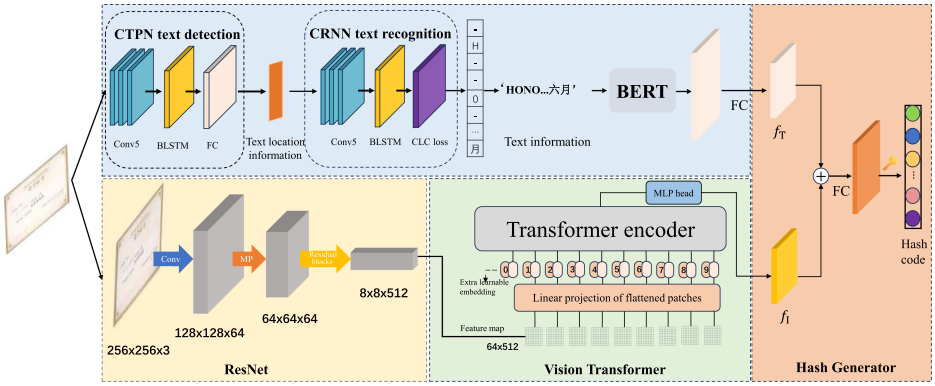


Fig. 3. Overview of the proposed model.

BERT model consists of two main tasks: the masked language model (MLM) and the next sentence prediction (NSP). The MLM task learns a rich linguistic representation by randomly masking some words in the input sentences and then allowing the model to predict these masked words. Additionally, the NSP task enables the model to understand the logical relationship between two sentences and trains the model by predicting whether two sentences are sequential.

III. Proposed Method

This section describes our proposed framework in detail. First, Section III-A introduces the overall architecture of the network, which consists of three main stages: text feature extractor, image feature extractor, feature fusion, and loss function detailed in Section III-B, III-C, III-D and III-E, respectively.

A. Overall Architecture

In this subsection, we present the ResViT-OCR hashing framework. Fig. 3 illustrates the proposed overall framework, which consists of a text-modal branch and an image-model branch, each including feature learning modules. The image and text are converted to a common coding space where similarities between image and text are preserved. The text features are converted to the feature size required for image modality using a fully connected (FC) layer, and then fused together. Finally, a hash sequence of length 50 is generated by an FC layer.

Specifically, we preprocess the training data to unify the image size and then feed it into the image and text feature extractors. The OCR module extracts the text content, and the extracted text information is input into BERT to generate text feature f_T. Similarly, the image is generated by the ResViT module to generate an image feature f_I. Then, f_T and f_I with the same size are fused to form the text image feature map. The hash generator converts the multidimensional feature map into a fixed-length image hash sequence. The effective combination of feature extractor and hash generator realizes the end-to-end learning process of the network model.

B. Text Extraction Module

The OCR module consists of two main parts: text region detection based on the Connectionist Text Proposal Network (CTPN) [31] and text recognition based on the Convolutional Recurrent Neural Network (CRNN) [32]. The CTPN combines the advantages of CNNs and Recurrent Neural Networks (RNNs) to generate high-quality text candidate regions. It further utilizes a regression network to pinpoint text box locations. The network architecture is shown in Fig. 4. Feature maps are extracted from the input image using CNNs, typically a pre-trained CNN (e.g. VGG or ResNet). The size of the feature map is $C \times H \times W$, where H and W denote the height and width of the image, and C is the number of channels of feature maps. On the last convolutional layer (Conv5) of the CNN, a spatial window of 3×3 size is used for dense sliding to generate the learned spatial features. These features are utilized to make predictions for multiple anchors, where the convolutional features of each window are $3 \times 3 \times C$. The obtained features are then input into a 256-dimensional bi-directional Long Short-Term Memory (BLSTM) Network, which captures the visual information of the image and provides the basis for subsequent text candidate generation. The output of the BLSTM network is input into the FC layer, and designs K vertical anchors to predict the y-coordinate of each proposal. The FC layer outputs three parts: $2K$ vertical coordinates represent the height of the selection box and the y-axis coordinates of the center; $2K$ scores represent the category information of the K anchors, stating whether they are characters or not; and K side-refinement offsets represent the horizontal offsets of the selection boxes. Then adjust the predicted anchors to determine the text candidate regions. Finally, output the accurate localized position of the text.

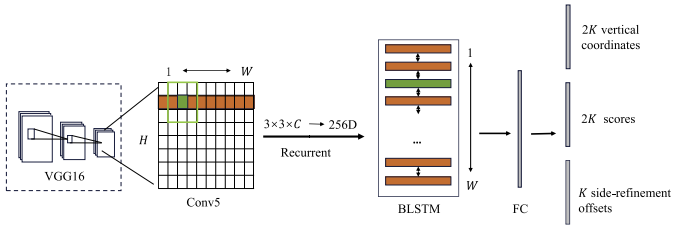


Fig. 4. Network architecture of the CPTN model. The green box indicates a 3×3 sliding window on Conv5, where each point is combined with the surrounding 3×3 area features to obtain a feature vector of length $3 \times 3 \times C$.

CRNN network for text recognition mainly consists of convolutional layers, recurrent layers, and transcription layer, as shown in Fig. 5. CNN is used to extract the features sequence

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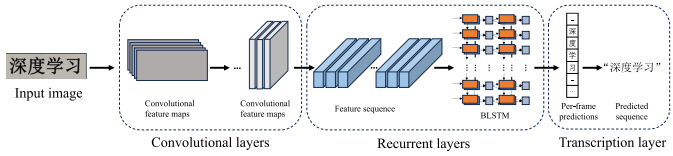


Fig. 5. Illustration of the CRNN framework for text recognition network. Convolutional layers: extract features from the input image and obtain a feature map. Recurrent layers: learn each feature vector in the sequence and output the predicted label distribution. Transcription layer: converts a series of label distributions obtained from the recurrent layer into a final labeled sequence.

of text location information obtained through CPTN. An RNN, such as an LSTM or a Gated Recurrent Unit (GRU), is used to capture the temporal dependencies in the sequence. This is crucial for understanding the context of the sequence. After RNN processing, each time step outputs a probability distribution of possible labels, including a special 'blank' label (usually denoted by '-' or a special null character). The main function of this blank label is to handle time steps without corresponding characters in the target sequence. In a transcription module that utilizes Connectionist Temporal Classification (CTC) technique, the CTC method evaluates all potential paths that can be taken to match the output of the model to the target sequence by inserting 'blanks' or repeating characters between characters. During decoding, the system removes repetitive and blank tags from the generated sequence to obtain the final text results. If it is assumed that the prediction results of each time frame are relatively independent, the conditional probability of each path can be expressed as

$$p(\boldsymbol{\pi}|\mathbf{y}) = \prod_{t=1}^{T} y_{\pi_t}^t, \tag{1}$$

where $\mathbf{y}$ and $\boldsymbol{\pi}$ represent the input sequence and valid paths, respectively, $y_{\pi_t}^t$ denotes the probability value on valid path π_t at time t, and T is the length of the sequence. By summing up the conditional probabilities of all valid paths, we can calculate the overall probability of the output string as

$$P(\mathbf{l}|\mathbf{y}) = \sum_{\boldsymbol{\pi}:\mathcal{B}(\boldsymbol{\pi})=\mathbf{l}} p(\boldsymbol{\pi}|\mathbf{y}), \tag{2}$$

where $\mathbf{l}$ and $\mathcal{B}$ represent the output string and the de-duplication operation, respectively. Finally, the character with maximum probability is taken as the output character for that time step to obtain the final predicted text result for the recognition of the whole text image.

Deep learning-based CRNN and CTPN models make full use of the advantages of CTPN in text region localization and the efficient performance of CRNN in sequence recognition. This scheme achieves the extraction and recognition of image text information and provides support for subsequent content authentication.

C. Image Hashing Module

In the document image hash module, the visual features of the document image are extracted, and a perceptual hash sequence is generated. We use ResNet and ViT in cascade to consider both local and global features. This process is mainly divided into two stages: local and global feature extractions.

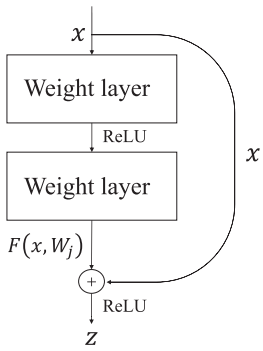


Fig. 6. Residual block internals. $F(x, W_j)$ is the residual function.

TABLE I
SPECIFIC NETWORK PARAMETERS FOR RESNET18

Layer-name	18-Layer
Conv1	7×7, 64, stride 2 3×3 maxpool, stride 2
Conv2	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$
Conv3	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$
Conv4	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$
Conv5	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$

We choose to use ResNet as the backbone network for local feature extraction. The problem of gradient vanishing with increasing network depth is addressed by introducing the concept of residual learning. ResNet is a deep convolutional network consisting of multiple residual blocks. Each block contains several convolutional layers, and the input is directly summed with the output through skip connections. This design facilitates gradient propagation and reduces the number of parameters. The ability to train very deep network structures improves the model's expressiveness and performance. Additionally, the residual connection preserves the original features, leading to a more stable learning process and enhancing both accuracy and generalization.

The internal structure of the residual block in ResNet is shown in Fig. 6. Each residual block contains two 3×3 convolutional layers, and each convolutional layer is followed by ReLU activation function. It can be represented as

$$z = F(x, W_j) + x, \tag{3}$$

where x and z are the input and output feature maps, respectively, W_j is the set of weight parameters for all operations in the j-th residual block. $F(x, W_j)$ represents the output of the layer with weights W_j . ResNet extracts features ranging from basic to complex, such as edges, textures, and object parts, thus efficiently capturing details and local structures in the image. This provides a solid foundation for further fusion of global

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features. We use ResNet18in our framework and Table I lists the specific network parameters of ResNet18.

ViT introduces the Transformer architecture to the vision domain, utilizing a self-attentive mechanism to capture global contextual information. Compared with ResNet, it is better at extracting global features. ViT segments an image into patches, which are treated as elements in a sequence and then processed by the Transformer's Encoder. The ViT model processes the local feature maps extracted by ResNet, which are first segmented into several non-overlapping patches of a fixed size $S \times S$ by ViT. These patches are then linearly expanded and converted into a series of tokens through a FC layer, with positional encoding added to preserve their spatial information. The transformed tokens are then fed into ViT's encoder for further processing. Inside the encoder, the self-attention mechanism allows the model to consider the information of all other tokens while processing each token. The three matrices $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are obtained by multiplying the tokens with three learnable matrices. Next, the model calculates the dot product of $\mathbf{Q}$ and $\mathbf{K}$ to evaluate the similarity between tokens, generating an attention scores matrix. A Softmax function processes this matrix to obtain a probability distribution, which is then multiplied with $\mathbf{V}$ to produce a weighted output. The model captures relationships between tokens from multiple perspectives by repeatedly applying the multi-head attention mechanism. The tokens, after being processed through multiple encoder layers, are eventually extracted. This mechanism allows the model to consider local features and capture global contextual information when processing images. Calculation of the attention mechanism can be represented as

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\mathrm{T}}}{\sqrt{D}}\right)\mathbf{V}, \tag{4}$$

where D represents the dimension of $\mathbf{K}$.

D. Feature Fusion

Before feature fusion, we use BERT to extract the textual feature of the sentence. We have chosen the only Chinese BERT model published by Google [33], because our training set and test set are Chinese document images. The BERT model is replaceable, and if we want to process English document images, we can choose other BERT models. BERT efficiently represents highly unstructured textual data as vectors, which is a trained stack of Transformer encoders. After extracting the textual information, we embed some special tags to distinguish each different sentence. At the beginning of the sentence, we embed '[CLS]' as the first token of each sequence, refer to the sequence of input tokens passed to BERT, and embed '[SEP]' at the end of the sentence. We can distinguish sentences by this unique token. Once the input tokens are ready, they are processed by each layer, which applies self-attention, passes the result through a feed-forward network, and hands it off to the next encoder. In terms of architecture, it remains the same as the Transformer. After extracting the text feature, it is fed into a FC layer, which means that the text and image features have the exact dimensions. Finally, the two features are spliced and fed into a FC layer to generate the final hash sequence.

E. Loss Function

During the training of the OCR module, the negative log-likelihood function is used as the loss function. Denote the training dataset by $X = \{\mathbf{I}_n, \mathbf{l}_n\}$, where $\mathbf{I}_n$ is the training images for the OCR module, and $\mathbf{l}_n$ is the truth label sequence. After the transcription layer in the CRNN module outputs the final text information, the CTC loss function is introduced in the training process.

$$\text{Loss}_{\text{text}} = -\sum_{\mathbf{I}_n,\mathbf{l}_n \in X} \log(\mathbf{l}_n|\mathbf{y}_n), \tag{5}$$

where $\mathbf{y}_n$ represents the input sequence produced by the CNN and RNN from $\mathbf{I}_n$.

When training the document image perception model, besides the original input image, denoted as $\mathbf{I}_{\text{ori}}$, the set of images input to the feature extractor consists of M perceptually identical images of $\mathbf{I}_{\text{ori}}$, denoted as $\widetilde{\mathbf{I}}_i, i = 1, 2, \ldots, M$, and the perceptually different images of $\mathbf{I}_{\text{ori}}$, denoted as $\mathbf{P}_i, i = 1, 2, \ldots, M$. Note that unlike the existing deep perception hashing methods, instead of perceptually different images, we here use tampered document images to increase the document tampering detection sensitivity of the proposed model in this paper. Each batch eventually obtains $2M + 1$ hash sequences, denoted as $\mathrm{H}(\mathbf{I}_{\text{ori}}), \mathrm{H}(\widetilde{\mathbf{I}}_1), \ldots, \mathrm{H}(\widetilde{\mathbf{I}}_M), \mathrm{H}(\mathbf{P}_1), \ldots, \mathrm{H}(\mathbf{P}_M)$, and each hash is with the length of L. Using these hash sequences, the corresponding distances of $\mathrm{H}(\mathbf{I}_{\text{ori}})$ and $\mathrm{H}(\widetilde{\mathbf{I}}_i)$, denoted as $\zeta_{\text{s}}^{(i)}$, are obtained for evaluating the feature mapping similarity between perceptually identical image pairs. Similarly, the corresponding perceptually different image pair distances of $\mathrm{H}(\mathbf{I}_{\text{ori}})$ and $\mathrm{H}(\widetilde{\mathbf{P}}_i)$, denoted as $\zeta_{\text{d}}^{(i)}$, also can be calculated. In this paper, these distances are measured by the mean square error (MSE), which measures the difference between the hash sequences of the comparison images and the original image hash sequences by calculating the mean of the squared differences, i.e.,

$$\zeta_{\text{s}}^{(i)} = \frac{1}{L}\sum_{k=1}^{L}\left(\mathrm{H}(\mathbf{I}_{\text{ori}})_k - \mathrm{H}(\widetilde{\mathbf{I}}_i)_k\right)^2, i = 1, 2, \ldots, M, \tag{6}$$

$$\zeta_{\text{d}}^{(i)} = \frac{1}{L}\sum_{k=1}^{L}\left(\mathrm{H}(\mathbf{I}_{\text{ori}})_k - \mathrm{H}(\widetilde{\mathbf{P}}_i)_k\right)^2, i = 1, 2, \ldots, M, \tag{7}$$

where L denotes the length of image hash, k denotes the k-th element.

In order to ensure that the hashing algorithm robustness and discrimination achieve a satisfactory balanced effect during the training process, the value of MSE is minimized by the gradient descent optimization algorithm, and the distances between pairs of images with the same perception and pairs of images with different perception is normalized by using the sigmoid function as a way to achieve faster convergence of the model. The loss function used is

$$\text{Loss}_{\text{hash}} = \alpha\frac{1}{M}\sum_{i=1}^{M}\mathrm{S}\left(\zeta_{\text{s}}^{(i)}\right) - \beta\frac{1}{M}\sum_{i=1}^{M}\mathrm{S}\left(\zeta_{\text{d}}^{(i)}\right), \tag{8}$$

where S is the sigmoid normalization function, and α, β are hyperparameters of the loss function that minimize the loss

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Algorithm 1: Training Process of Robust Perceptual Document Image Hashing.

Input: Training and validation document images, the proposed ResViT-OCR model $\mathbf{H}(\cdot)$, epoch
Output: Hash sequence $\mathrm{H}(\cdot)$
Each batch contains the original image $\mathbf{I}_{\text{ori}}$, the perceptually similar images $\widetilde{\mathbf{I}}_i$, and the tampered images $\mathbf{P}_i$, $i = 1, 2, \ldots, M$;
Initialize the parameters of $\mathbf{H}(\cdot)$;
$t = 0$;
while $t <$ epoch **do**
    for each batch **do**
        Generate hash $\mathrm{H}(\cdot)$ through $\mathbf{H}(\cdot)$
        Calculate hashing distances between corresponding image pairs, i.e., $\text{Loss}_{\text{hash}}$, via (8);
        Update the model parameters of $\mathbf{H}(\cdot)$ to minimize $\text{Loss}_{\text{hash}}$;
    end for
    Feed validation images into the network to calculate hash values for evaluation;
    $t = t + 1$;
end while
return $\mathrm{H}(\cdot)$

function to generate hash distances that can distinguish between similar and different images.

To better understand, Algorithm 1 presents the training process of the proposed robust perceptual document image hashing.

IV. Experimental Results

In this section, we first describe the datasets and training parameter settings. In order to evaluate the effectiveness of the proposed model, we conduct a series of experiments, including robustness, discriminative ability, discussion of threshold, authentication ability, adversarial attacks, ablation experiments, computational complexity, and key-dependent performance.

A. Experiment Setup

1) Dataset Construction: We construct a text tampering dataset to evaluate the proposed scheme's effectiveness and superiority. We employ the tampering method from [34], which selectively tampers to generate realistic images of tampered documents. In [34], it consists of selective copy-paste and selective generation. The former acquires groups of texts with similar styles and performs copy-paste in the grouped text instances to generate the tampered texts. The latter uses OpenCV [35] to erase their original text content and then prints the new text with a predefined similar style and font. Following [34], the entire data synthesis approach for text tampering we used is described as follows: (1) use powerful open-source OCR tools such as PPOCRLabel in Paddle-OCR [36] to obtain the bounding boxes of the words and characters and extract the parts of the text that we want to tamper; (2) use styleText [3] to separate the foreground color of the text from its background and use the background to generate new text content; (3) selectively copy and paste the generated new text content to the original image to obtain the tampered document image.

We randomly select 500 document images as original images from the Document Forgery Attack dataset of Alibaba Tenchi Competition [37]. Using the method described above, we tamper with these images so that each original image yields 35 corresponding tampered versions. Based on this, we perform multiple content-preserving operations with different parameters on each image, as listed in Table II. We use seven types of image content-preserving operations, including JPEG compression, Gaussian noise, cropping, and scaling. These adjustments to the document images do not affect the semantic information they contain; hence, we all consider the images to be perceptually similar. Therefore, each original image has 35 perceptually similar versions, and the proposed document image perceptual hashing dataset for training, referred to as DIPHD1, consists of $500 + 500 \times 35 + 500 \times 35 = 35,500$ images. In order to ensure the diversity of the detection data, we also choose DocuImg, a document image dataset proposed by Chen et al. [38] as the detection database for the following experiments.

2) Implementation and Training Details: The framework proposed in this paper is implemented in Pytorch and trained on an NVIDIA RTX3090 GPU to obtain the final hash model. We train the OCR module and the document image perceptual hashing module in steps. The Synthetic Word dataset [39] is used during the training of the OCR module, which is widely used in computer vision and machine learning, especially in text recognition and detection tasks. This dataset consists of synthetically generated images, each of which is automatically generated and contains text in different fonts, sizes, colors, and backgrounds to cover a wide range of possible scenarios. For the training of the OCR module, we use RMSProp [40] as the optimizer and set the learning rate as 1×10^{-4}. The epoch of the model is set to 200.

In training the document image perceptual hash model, we use our proposed and constructed tampered document image dataset DIPHD1. This dataset includes the original document images, the similar version of the original images, and the tampered version of the original images. In addition, we set the training learning rate to e^{-4}, optimize the algorithm using the Adam optimizer [41], and the corresponding batch size is set to 71. Respectively, the model's epoch is set to obtain the final perceptual hash network. Before feeding the images into the model, we normalized the images to a size of 224×224 for model training. In addition, the length of the hash sequence is set to 50.

3) Similarity Measurement: We use MSE to calculate the distance between two image hashes $\mathrm{H}(\mathbf{I})$ and $\mathrm{H}(\mathbf{I}')$ as

$$\mathrm{D}\left(\mathrm{H}(\mathbf{I}), \mathrm{H}(\mathbf{I}')\right) = \frac{1}{L} \sum_{k=1}^{L} \left(\mathrm{H}(\mathbf{I})_k - \mathrm{H}(\mathbf{I}')_k\right)^2, \tag{9}$$

where $\mathrm{H}(\mathbf{I})_k$ and $\mathrm{H}(\mathbf{I}')_k$ denote the k-th element of $\mathrm{H}(\mathbf{I})$ and $\mathrm{H}(\mathbf{I}')$, respectively. If the similarity of the two hashes is less than a preset threshold, their corresponding images are considered to be perceptually similar; conversely, they are considered to be tampered images.

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TABLE II
SEVEN DIFFERENT CONTENT-PRESERVING OPERATIONS WITH DIFFERENT PARAMETER SETTINGS

Operation	Parameter	Values
JPEG compression	Quality factor	30, 50, 70, 90, 95
Gaussian noise addition	Variance	0.01, 0.02, 0.03, 0.04, 0.05
Gaussian blur	Kernel size	1, 3, 5, 7, 9
Motion blur	Kernel size	1, 3, 5, 7, 9
Rotation + Cropping	Rotation angle	1,2,3,4,5
Gamma correction	Gamma coefficient	0.50, 0.70, 0.90, 1.10, 1.30
Scaling	Scaling ratio	0.9,1.1,1.3,1.5,2.4

TABLE III
STATISTICS OF HASH DISTANCES BASED ON CONTENT-PRESERVING IMAGE OPERATIONS

Operation	Minimum	Maximum	Mean	Standard Variation
JPEG compression	0.1339	11.7247	1.6460	1.5930
Gaussian noise addition	0.1964	13.5843	1.5616	1.5698
Gaussian blur	0.2158	9.6035	1.7812	1.6825
Motion blur	0.1953	14.7848	1.8626	1.8649
Rotation+Cropping	0.1594	12.6651	1.6813	1.6677
Gamma correction	0.1677	15.2626	1.5287	1.6148
Scaling	0.1964	14.5099	1.7334	1.8157

B. Perceptual Robustness

In order to verify the perceptual robustness of the proposed algorithm to the content-preserving operations, additional 100 images are randomly selected from the Alibaba Tenchi Competition dataset as the original version and use the content-preserving operations on them as shown in Table II. Each image generates 35 perceptually similar images, and 35 tampered images are generated using the tampering method in dataset construction. Denote this test dataset as DIPHD2. In this experiment, only the original images and their perceptually similar versions are used. Then, we calculate the similarity distance between the original image and the perceptually similar image hash sequence according to (9).

We refer to each original image and its perceptually identical versions as an image group. The total number of hash distances between perceptually identical pictures are $100 \times 35 + 100 = 3,600$ pairs of hash distances. Table III shows the statistics of hash distances for dataset DIPHD2 based on the different content-preserving operations. The hash distances range from 0.0000 to 15.2626, the mean of these distances is 1.6850, and the standard deviation is 1.6869. Table III shows that the minimum distances for all content-preserving operations are close to 0, and the average hash distances are almost always less than 2. However, the maximum hash distances under the Gamma correction and motion blur operations are relatively large because the accuracy of OCR technology relies heavily on the sharpness of the input image. The accuracy of the OCR technique is highly dependent on the clarity of the input image. The OCR technique may have difficulty accurately recognizing blurry, too bright documents, or documents that contain complex layouts, leading to relatively large hash distances when the parameter values are set relatively large under some operations. However, most perceptually similar images are correctly recognized and can be distinguished from tampered images, which will be discussed in more detail in Section IV-D.

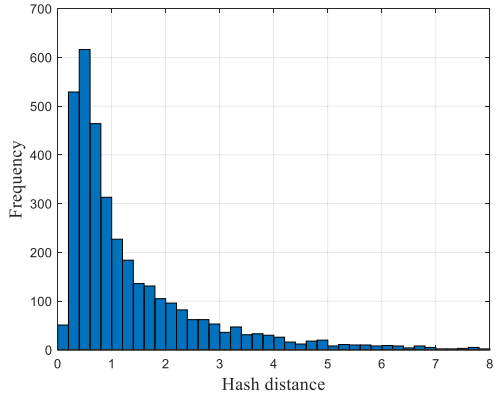


Fig. 7. Histogram of the hash distance between 3,500 sets of perceptually identical images obtained from 100 original test images in DIPHD3.

In order to further verify the robustness of the scheme, we also randomly select 100 images from the DocuImg dataset as the original versions, referred to as dataset DIPHD3, and conduct different content-preserving operations listed in Table II. Fig. 7 illustrates the histogram of 3,500 hashing distances between perceptual identical images. From the figure, we can conclude that the hash distances under all content-preserving operations are mainly concentrated in the range of $[0, 2)$, and even the minimum values of hash distances are close to 0, which indicates that the hash values are almost the same as those processed version and the original images. Based on the above experiments, we can conclude that our document image hashing scheme shows satisfactory robustness to the commonly used content-preserving operations.

C. Discrimination Capability

A robust document image hashing scheme needs to be discriminative, and for tampered images, the hash value should be significantly different from the original image. To verify

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Fig. 8. Some tampered document images.

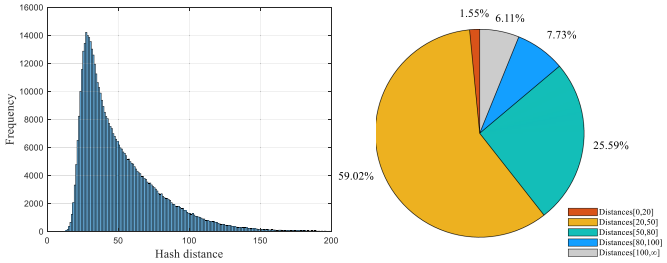


Fig. 9. Distance distribution between perceptually distinct images.

the discrimination capability of this scheme, additional 100 images are randomly selected from the DocuImg dataset, and 900 tampered images are generated for testing by the tampering method described in dataset construction. Denote this dataset as DIPHD4, and some tampered examples are shown in Fig. 8. Then the hash value for each image in DIPHD4 is generated. We calculate the hash distances between each image and the other 999 images separately and obtained a total of 499,500 hash distances. The mean value of these hash distances is 50.69, and the standard deviation is 26.88. In order to show the distribution of these hash distances more intuitively, the experiments are statistically analyzed using pie charts, and the specific results are shown in Fig. 9. The histogram and pie chart show that the hash distances of the perceived different images are mainly distributed in the interval $[20, +\infty)$. Compared with the distribution of hash distances of the images of the content-preserving operations in the robustness experiments mentioned in Section IV-B, it can be found that there is a significant difference between the two in the range of hash distances. This indicates that the proposed hashing network is able to yield significant differences between the hash values of different images and thus has a good discriminative ability.

D. Performance of Content Identifications

In order to better evaluate the superiority of our scheme in content authentication, authentication experiments are conducted using the constructed dataset DIPHD2, which contains original images, perceptually similar images and tampered images. Fig. 10 illustrates the proposed content recognition workflow. We extract the hash values of the test images using the proposed algorithm and then calculate their hash distance using MSE. If the distance is smaller than a preset threshold T, the proposed framework considers that the document image has not been tampered with; otherwise, it is judged as tampered or a different image. Our algorithm is compared with the current state-of-the-art perceptual image hashing algorithm schemes DAE [22], MCNN [10], ProtoHash [42], TMCIH [43], Swin-T [23] as well as the traditional methods RP-IVD [44] and 2D-2D PCA [45]. It is worth noting that the robustness and discriminative power of hashing algorithms often conflict, and an increase in the power of one affects the performance of the other. We use ROC curves to better demonstrate the algorithm's authentication ability. The True Positive Rate (TPR) and False Positive Rate (FPR) are calculated by setting different thresholds as

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{10}$$

and

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{TN} + \mathrm{FP}}, \tag{11}$$

respectively, where TP is the number of perceptually similar images that are correctly judged to be similar, FN is the number of perceptually similar images that are incorrectly judged to be different, FP is the number of perceptually distinct images that are incorrectly judged to be similar, and TN is the number of perceptually distinct images that are correctly judged to be different. TPR and FPR represent the robustness and discriminatory ability of the hashing algorithm, respectively. The closer the ROC curve is to the upper left corner, the better the performance of content authentication achieved by the scheme. The ROC curves in Fig. 11 demonstrate the content authentication performance of various hashing schemes. It can be observed that our scheme is closer to the upper left corner than the other schemes because we have added the OCR module to utilize the text information to help the hashing algorithms accomplish the content authentication task. The contrasting hashing algorithms perform well when dealing with masked and background-tampered images, but perform poorly in scenarios where textual information tampering is recognized. This finding further shows the effectiveness of the proposed scheme in this paper in solving the content authentication of document images.

The F_1 score is a measure for evaluating algorithm through combining the accuracy and recall of an algorithm's classification. The maximum value of the F_1 score is 1, and the minimum value is 0. A higher F_1 score indicates better classifier performance. To more fully evaluate the content authentication capabilities of each hashing algorithm, we calculate the F_1 scores of various algorithms for different content-preserving operations as

$$F_1 = \frac{\mathrm{TP}}{\mathrm{TP} + \frac{1}{2}(\mathrm{FP} + \mathrm{FN})}. \tag{12}$$

Table IV shows the comparison of F_1 scores for different image processing operations, where our scheme is closer to 1 and outperforms all other schemes for all content operations. This is because in detecting document images, the OCR technique of our scheme can provide more accurate and intelligent text detection and recognition capabilities, especially in the face of text tampering with only a tiny portion of the text, and can provide more efficient, accurate, and robust text recognition services. Therefore, the image hash generated by our proposed method shows good authentication capability in document image tampering detection, validating the algorithm's effectiveness.

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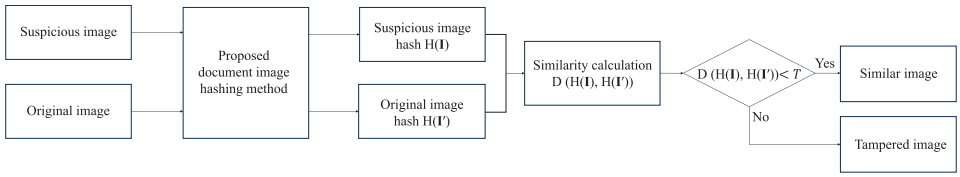


Fig. 10. Workflow of image content authentication.

TABLE IV
COMPARISON OF F_1 SCORES OF VARIOUS HASHING ALGORITHMS FOR DIFFERENT CONTENT-PRESERVING OPERATIONS

Operation	RP-IVD	2D-2D PCA	DAE	MCNN	ProtoHash	TMICH	Swin-T	Ours
JPEG compression	0.8781	0.9023	0.8224	0.7785	0.9223	0.9270	0.7752	**0.9397**
Gaussian noise addition	0.0747	0.2529	0.7984	0.6676	0.7324	0.6645	0.6676	**0.9397**
Gaussian blur	0.7769	0.5406	0.8243	0.9160	0.9419	0.5848	0.7523	**0.9546**
Motion blur	0.4646	0.4222	0.7104	0.6782	0.8220	0.6657	0.6668	**0.9498**
Rotation+Cropping	0.0168	0.0078	0.2532	0.6721	0.7475	0.6659	0.6670	**0.9399**
Gamma correction	0.7583	0.0378	0.8206	0.6675	0.8033	0.6504	0.6667	**0.9314**
Scaling	0.8784	0.8558	0.8220	0.9165	0.9393	0.6667	0.7715	**0.9641**

The best results are in bold.

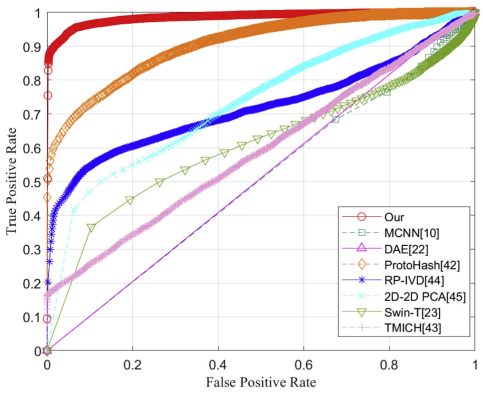


Fig. 11. ROC curves of content authentication performance for the proposed scheme, [22], [10], [42], [44], [45], [43], and [23].

E. Threshold Discussion

There is a trade-off between the robustness and discriminability of document image hashing schemes, where a decrease in the other inevitably accompanies an increase in the performance. A smaller threshold value ensures better discriminative capability but at the expense of robustness. Therefore, choosing an appropriate threshold is of great importance. For optimal performance, the threshold value corresponding to the point closest to the upper left corner of the ROC curve is recommended. In different application scenarios, users may assign different levels of importance to robustness or discriminability. Table V provides the authentication correctness results for perceptually similar images and tampered images, where all selected images are from DIPHD2 and Γ_i and Γ_d denote the accuracy of perceived robustness and discriminative ability under different thresholds T, respectively. Taking the threshold value of 1.5 as an example, in this case, 92.10% of the same images are correctly judged as perceiving the same, and 96.4% of the tampered images are correctly judged as perceiving different. It can be found that when the threshold T is 2.1, 93.54% of similar images are correctly judged as perceptually identical, 95.46% of tampered images are correctly judged, and the average of the correctness of perceptual robustness and discriminative ability is largest. This threshold corresponds to the point closest to the upper left corner on the DIPHD2-based ROC curve of this paper's scheme. To maintain a balance between robustness and discrimination, we recommend a threshold T of 2.1.

F. Ablation Experiments

In order to verify the impact of OCR module on the performance of content authentication in document image hashing schemes, we design ablation experiments and compare the performance by ROC curves. In this experiment, a complete hash model containing the OCR module is first set as the baseline model. Subsequently, an ablation model is created by removing the OCR module, leaving all other processing steps unchanged, and the two models are called IOH and IH, respectively. This facilitates us to directly observe the change in the performance of the model in terms of content authentication of document images when the OCR module is not used, and thus to evaluate the performance of the model in recognizing and authenticating the content of document images in the absence of the OCR technique for text extraction performance. Note that each model is tested on DIPHD2 In this experiment.

The performance comparison with or without the OCR module is visualized in Fig. 12. The ROC curve of the hashing algorithm, which includes the OCR module, is higher than that of the ablation model, showing a tendency to be closer to the upper left corner, which indicates that the model which includes the OCR module is able to achieve a higher true positive rate. A more accurate authentication of the content of the document image while maintaining a low false positive rate. By analyzing the ROC curves, it can be seen that the OCR module significantly contributed to improving the accuracy and reliability of the document image hashing scheme for the content authentication task. In order to further illustrate the impact of the OCR module on the detection algorithm, we calculate the hash distances distribution with and without the OCR module, using the

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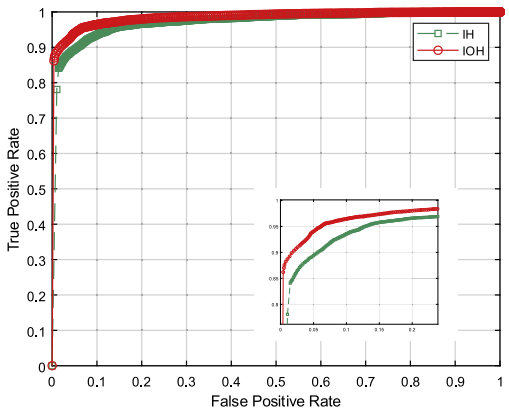


Fig. 12. ROC curves with or without OCR module.

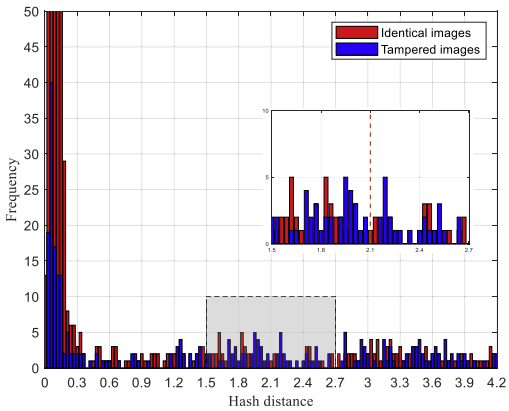


Fig. 13. Histogram of the hash distance between the identical images and tampered images without the OCR module.

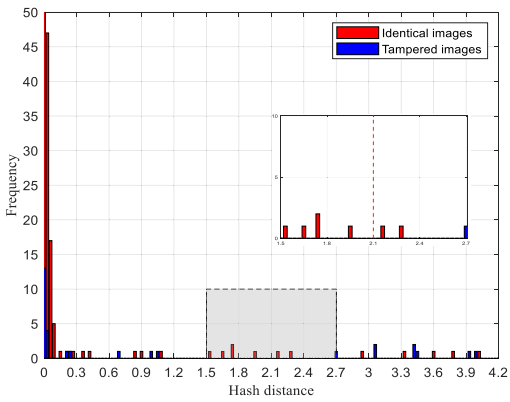


Fig. 14. Histogram of the hash distance between the identical images and tampered images with the OCR module.

DIPHD2 dataset. Figs. 13 and 14 show the hash distances of samples at the threshold edge. Many tampered images are incorrectly classified as perceptually similar without an OCR module. This confirms the important role of the OCR module in improving the document image content authentication model.

G. Adversarial Attacks

To test the adversarial security of proposed scheme, we used FGSM [46] and PGD [47] to attack the perceptual image hash module. We selected 100 images from the DocuImg dataset and performed adversarial attacks on these 100 images using PGD and FGSM methods to test the accuracy. We set different perturbation threshold parameters ε, and each image generates 100 adversarial samples. We validate these 1000 images, and Fig. 15 shows the comparison of the model's performance under different perturbation thresholds for the adversarial example methods, FGSM and PGD. We calculate the accuracy of the model's correct prediction based on the threshold $T = 2.1$. When the hash distance is less than the threshold, it indicates that the adversarial sample is still correctly judged, proving that the attack was unsuccessful, thereby assessing our model's defense capability.

From the attack results, it can be seen that our model has a certain degree of defense capability. Under the attack of FGSM and PGD, when the perturbation threshold is less than 0.025, these adversarial sample images can be correctly distinguished as perceptually similar images. For the FGSM method, when the perturbation threshold is greater than 0.3, the accuracy decreases to 59.6%. For the PGD method, when the perturbation threshold is greater than 0.35, the accuracy drops to 62.2%. Comparing the attack results without the OCR module, it can be concluded that adding the OCR module has improved the defense performance of the model.

TABLE V
RESULTS OF Γ_i, Γ_d AND THEIR AVERAGE UNDER DIFFERENT THRESHOLDS

Threshold T	Γ_i	Γ_d	$(\Gamma_i + \Gamma_d)/2$
1.5	0.9210	0.9640	0.9425
1.6	0.9233	0.9623	0.9428
1.7	0.9255	0.9605	0.9430
1.8	0.9376	0.9589	0.9433
1.9	0.9302	0.9571	0.9437
2.0	0.9329	0.9557	0.9443
2.1	0.9354	0.9546	**0.9500**
2.2	0.9373	0.9534	0.9454
2.3	0.9389	0.9517	0.9453
2.4	0.9399	0.9503	0.9451
2.5	0.9414	0.9488	0.9451

H. Computational Complexity

Computational complexity plays a crucial role in practical applications. In this section, we evaluate the time complexity of our scheme and compare it with DAE [22], MCNN [10], Proto-Hash [42], TMCIH [43], Swin-T [23] as well as RP-IVD [44] and 2D-2D PCA [45]. We randomly select 10,000 images from DocuImg dataset and resize them to the size of 256×256 using a bilinear interpolation. We count the time spent by each hashing algorithm to generate 1, 10, 100, 1,000, and 10,000 image hashes. By calculating the average of these times, the time taken by the algorithms to generate hashes can be obtained. It is worth noting that all the algorithms generate hash sequences with a length of 50 digits, except for the two conventional methods, RP-IVD and 2D-2D PCA. The average time taken by each algorithm to generate hash values is shown in Table VI. Based on the table, although the proposed model is slightly slower than other deep hashing schemes; this is probably due to the fact that

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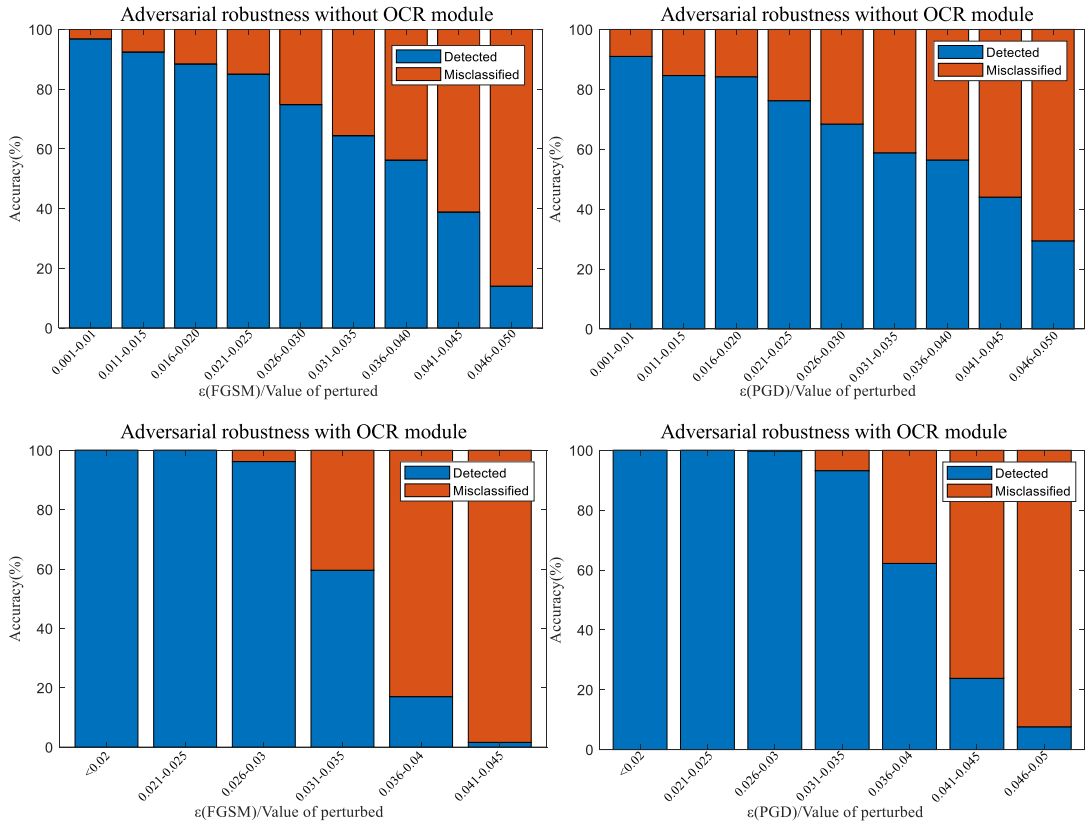


Fig. 15. Performance of adversarial robustness under FGSM and PGD attacks.

TABLE VI
AVERAGE TIME FOR GENERATING HASH SEQUENCES WITH DIFFERENT ALGORITHMS

Algorithm	Average time	Hash length
RP-IVD	283.24ms	40 digits
2D-2D PCA	61.40ms	32 digits
DAE	67.7ms	50 digits
MCNN	22.68ms	50 digits
ProtoHash	38.45ms	50 digits
TMICH	21.5ms	50 digits
Swin-T	38.9ms	50 digits
Ours	188.07ms	50 digits

the hash sequences generated by our model need to extract the textual content first and then use the BERT model to change the textual content into feature vectors. The BERT model possesses many parameters, requiring more computational resources and time to process the input data. Therefore, in the subsequent research, we will continue to optimize the model to improve computational efficiency.

I. Key Dependence

Key dependence is essential to ensure the hash function's efficiency, security, and reliability. This requires the hash algorithm to generate secure hashes while ensuring excellent robustness and differentiation. In our scheme, key H_{key} is used for the final stage of hash generation. In the validation experiments,

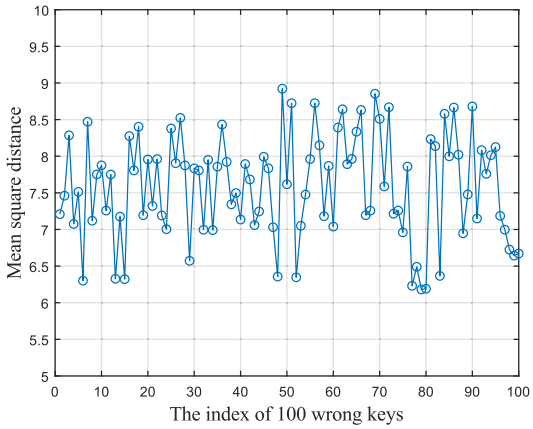


Fig. 16. Distance between hashes generated from different keys.

we generate random keys to encrypt the image hash, and the whole framework does not change the parameters other than the key. To validate the key dependency of the proposed framework, we select an image from the DocuImg dataset, choose a key to generate an encrypted hash key H_{key} for it, and then randomly select 100 incorrect keys to obtain other encrypted versions. The key security performance is quantitatively analyzed by calculating the mean square error distance between these 100 incorrect key encryption hashes and H_{key}. The experimental results are presented in Fig. 16, where the horizontal axis represents the index encrypted by the 100 different keys, and the vertical axis

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shows the distance between H_{key} and the sequence of incorrect hashes.

From the experimental results, it can be observed that there is a significant difference in the distance between the hashes generated by the correct key and the wrong key. Specific values show that the maximum value of these distances reaches 8.92, the minimum value is 6.18, and the average value is 7.59, accompanied by a standard deviation of 0.70. It is worth noting that even the smallest hash distance far exceeds the preset threshold in this paper, which means that the hash sequence cannot be accurately obtained or recovered without the correct key. This experimental result verifies that the hashing framework in this chapter has a high key dependency and demonstrates satisfactory security.

V. Conclusion

In this paper, we propose a perceptual hashing algorithm for document images. The whole algorithm is divided into OCR modules and text image hash modules, and they are trained separately to compose a complete hashing framework. The text information recognized by OCR is used to detect the tampering of characters in the document image, while the perceptual image hash module is used to detect the tampering of visual features such as background and masking of the image. Combining the OCR system with perceptual image hashing technology, it realizes all-around content authentication of document images, taking into account both textual and visual features of document images. In addition, by adopting a deep residual network with ViT cascade, we realize the comprehensive extraction of visual features of the image. In order to verify the effectiveness of the proposed algorithm, we have designed a series of experiments. The experimental results show that the proposed perceptual hashing scheme exhibits excellent performance in the face of tampering detection of document images. Experimental comparisons with other state-of-the-art schemes and results based on ROC and F_1 score evaluation metrics show that our algorithm also achieves satisfactory results in terms of robustness and discrimination.

We will continue improving tamper detection performance under OCR noise in future research. In addition, we will further improve the proposed scheme by exploring more effective fusion methods to integrate text and image feature information, simplifying the model complexity by utilizing knowledge distillation and quantization, and developing more lightweight modules to enhance computational efficiency.

Acknowledgment

The authors would like to thank the anonymous reviewers for their valuable suggestions, which helped to improve this paper. Our dataset can be downloaded from https://1drv.ms/f/c/f6dba7ae8a6cd6de/Es-D-F6fM7VEnIKcvcqj-3MBRU0SZ39a0qtvUzLPBO-Ahw?e=7vg153. The code of this paper is available at https://github.com/HengYao/Document-Image-Perceptual-Hashing-for-Content-Authentication.

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